PEDIATRIC MEDICINE · LECTURE 5 · QUESTION BANK

Rheumatic Fever

الحمى الروماتيزمية — بنك الأسئلة

70	6	12
Past exam questions	NCLEX-style	Flashcards

المصادر اللي اتجمعت منها أسئلة المحاضرة دي:

أسئلة سابقة · ميدترم 2022 (أطفال نظري) · ميدترم 2024 (ترم أول 2023-2024) (أطفال نظري)

جنب كل سؤال مصدره. لو السؤال اتكرر في كذا امتحان مكتوب كل مرة بمصدرها، وعليه علامة "تكرر". الإجابات كلها في آخر الملف.

اللَّهُمَّ لا سَهْلَ إِلَّا ما جَعَلْتَهُ سَهْلًا، وأنتَ تَجْعَلُ الحَزْنَ إذا شِئْتَ سَهْلًا — صلِّ على النبي ﷺ

Part A · Past Exam Questions

أسئلة الدفعات السابقة

1 أسئلة سابقة

What is the primary definition of Rheumatic Fever?

a. A viral infection in children
b. An inflammatory disease following a bacterial infection
c. A hereditary condition
d. An immune deficiency disorder

2 أسئلة سابقة

Which of the following is the main causative agent of Rheumatic Fever?

a. Streptococcus pyogenes
b. Escherichia coli
c. Staphylococcus aureus
d. Clostridium difficile

3 أسئلة سابقة

What are the most common symptoms of Rheumatic Fever?

a. Sore throat and difficulty swallowing
b. Fever, joint pain, and skin rash
c. Nausea and vomiting
d. Severe headache and dizziness

4 أسئلة سابقة

According to the Jones Criteria, which of the following is a major criterion for diagnosing Rheumatic Fever?

a. Subcutaneous nodules
b. Fatigue
c. Cough
d. Chest pain

5 أسئلة سابقة

Which investigation is most useful for confirming a recent streptococcal infection?

a. Throat swab culture
b. Chest X-ray
c. Echocardiography
d. Complete blood count

أسئلة سابقة

6 What is the primary goal of Rheumatic Fever treatment?

a. Relieving symptoms only
b. Preventing recurrence and complications
c. Improving physical fitness
d. Controlling blood pressure

أسئلة سابقة

7 What is a common complication of Rheumatic Fever affecting the heart?

a. Valvular damage
b. Myocardial infarction
c. Pericardial effusion
d. Atrial fibrillation

أسئلة سابقة

8 The mainstay of prevention for Rheumatic Fever is:

a. Vaccination
b. Early treatment of streptococcal infections
c. Routine physical check-ups
d. Antibiotic prophylaxis

أسئلة سابقة

9 Which medication is commonly used for long-term prophylaxis in Rheumatic Fever?

a. Penicillin
b. Paracetamol
c. Amoxicillin
d. Azithromycin

أسئلة سابقة

10 Rheumatic Fever most commonly affects which age group?

a. Infants under 1 year
b. Children aged 5-15 years
c. Adults aged 30-50 years
d. Elderly over 65 years

أسئلة سابقة

11 Which major organ system is primarily involved in Rheumatic Fever?

a. Nervous system
b. Cardiovascular system
c. Respiratory system
d. Digestive system

أسئلة سابقة

12 Which of the following is not part of the Jones Criteria?

a. Arthritis
b. Chorea
c. Skin ulcers
d. Carditis

أسئلة سابقة

13 What is the gold standard diagnostic test for Rheumatic Heart Disease?

a. Echocardiography
b. Electrocardiogram (ECG)
c. MRI
d. CT scan

أسئلة سابقة

14 Which is a minor criterion in the Jones Criteria?

a. Fever
b. Carditis
c. Chorea
d. Subcutaneous nodules

أسئلة سابقة

15 Which clinical feature suggests Sydenham's chorea in Rheumatic Fever?

a. Uncontrollable jerky movements
b. Severe chest pain
c. Shortness of breath
d. Muscle weakness

أسئلة سابقة

16 The pathogenesis of Rheumatic Fever primarily involves:

a. Autoimmune cross-reactivity
b. Direct bacterial invasion
c. Viral mutation
d. Nutritional deficiencies

أسئلة سابقة

17 Which antibiotic is preferred for treating streptococcal throat infections to prevent Rheumatic Fever?

a. Erythromycin
b. Penicillin
c. Ciprofloxacin
d. Metronidazole

أسئلة سابقة

18 What is the main sign of acute carditis in Rheumatic Fever?

a. Heart murmur
b. Edema
c. Severe hypertension
d. Cyanosis

أسئلة سابقة

19 How long should antibiotic prophylaxis continue in patients with Rheumatic Fever without carditis?

a. 1 year
b. 5 years
c. 15 years
d. Until age 21

أسئلة سابقة

20 Rheumatic fever is caused by Streptococcus pyogenes. (True / False)

a. True
b. False

أسئلة سابقة

21 Carditis is a minor criterion in the Jones criteria. (True / False)

a. True
b. False

أسئلة سابقة

22 Rheumatic fever primarily affects the respiratory system. (True / False)

a. True
b. False

أسئلة سابقة

23 Antibiotic prophylaxis is unnecessary if a patient has no symptoms of rheumatic fever. (True / False)

a. True
b. False

أسئلة سابقة

24 Subcutaneous nodules are a major criterion in the Jones criteria. (True / False)

a. True
b. False

أسئلة سابقة

25 Symptoms of rheumatic fever typically appear how long after a streptococcal infection?

a. 1–2 days
b. 3–5 days
c. 2–3 weeks
d. 1 month

أسئلة سابقة

26 Which joints are most commonly affected by pain or swelling in rheumatic fever?

a. Shoulders and hips
b. Knees, ankles, elbows and wrists
c. Fingers and toes
d. Spine and neck

27 What type of joint pain is often seen in rheumatic fever? أسئلة سابقة

a. Persistent pain in one joint
b. Pain that moves from joint to joint (migratory)
c. Pain relieved by rest
d. Pain without swelling or redness

28 Which is NOT a typical manifestation of rheumatic fever? أسئلة سابقة

a. Fever
b. Joint swelling
c. Skin rash
d. Headache

29 Under the 2015 Modified Jones Criteria, which symptom most likely indicates rheumatic fever? أسئلة سابقة

a. Severe headache
b. Hot or tender joints
c. Abdominal pain
d. Nausea and vomiting

30 Which body systems are typically affected in rheumatic fever? أسئلة سابقة

a. Respiratory, digestive and immune
b. Musculoskeletal, cardiovascular and nervous
c. Digestive, endocrine and respiratory
d. Cardiovascular, digestive and reproductive

31 Under the 2015 Modified Jones Criteria, which is a common sign? أسئلة سابقة

a. Skin lesions
b. Fever and joint pain
c. Gastrointestinal distress
d. Shortness of breath

32 Which is the characteristic skin manifestation of rheumatic fever? أسئلة سابقة

a. Erythema nodosum
b. Erythema marginatum
c. Psoriasis
d. Vitiligo

33 What is the primary treatment of rheumatic fever (eradicating the streptococcus)? أسئلة سابقة

a. Corticosteroids
b. Aspirin
c. Penicillin
d. Sulfadiazine

34 Which is NOT a sign or symptom of rheumatic fever? أسئلة سابقة

a. Joint pain
b. Fatigue
c. Chest pain
d. Nausea

35 One minor manifestation for rheumatic fever is أسئلة سابقة

a. Erythema marginatum
b. Poly arthritis and chorea
c. Increased sedimentation rate
d. Painless subcutaneous nodules

36 A nurse is preparing child for discharge after diagnosed with rheumatic fever that is the following instruction

أسئلة سابقة

a. Only give penicillin for dental work
b. Give aspirin for management chorea
c. Give penicillin until 21 years old
d. Give penicillin for one month total

37 Which of the following nursing diagnosis is most appropriate for child with rheumatic fever?

أسئلة سابقة

a. High risk for injury
b. Imbalance munition more than requirement
c. Delayed growth and development
d. Impaired gas exchange

38 In case of rheumatic heart fever with Myocarditis, the chest x-ray reveals normal heart size

أسئلة سابقة

a. True
b. False

39 Rheumatic fever is inflammation of the joints, heart, skin, and nervous system that occurs in children as a complication of untreated streptococcal infection of the throat. While most people with rheumatic fever recover, which part of the body may be permanently damaged in a small percentage of people?

أسئلة سابقة

a. Digestive tract
b. Ears
c. Heart
d. Lungs

40 Which of the following is usually the first symptom of rheumatic fever?

أسئلة سابقة

a. Chest pain or palpitations
b. Jerky, uncontrollable movements
c. Joint pain and fever
d. Rash

41 Penicillin is safely given to patients by the IV route. (True / False)

أسئلة سابقة

a. True
b. False

42 Each of the following drugs is used to treat rheumatic fever in children EXCEPT:

أسئلة سابقة

a. Acetaminophen
b. Antibiotics
c. Anti-inflammatory drugs
d. Corticosteroids

43 Patient with polyarthritis and carditis without evident of CHF (congestive heart failure) is treated with

أسئلة سابقة

a. IV salicylates
b. Oral salicylates
c. Corticosteroid
d. Both B and C

44 Which symptom presented is NOT a sign of rheumatic fever?

أسئلة سابقة

a. Truncal rash
b. Joint tenderness
c. Changes in vision
d. Nausea

45 Rheumatic fever is caused by which of the following bacteria?

أسئلة سابقة

a. Pseudomonas
b. Staphylococcus
c. Streptococcus
d. Shingella

46 Which of the following factors in the patient's history and physical is LEAST relevant to the diagnosis of Rheumatic fever? أسئلة سابقة

a. Sore throat from 3 weeks previous
b. Living in the current conditions of Flint, MI
c. The mother's history of type-2 diabetes
d. A possible genetic factor from the father who's history is unknown

47 The provider suspects rheumatic fever and orders blood work. Which results indicate inflammation and cardiac involvement? (Select all that apply) أسئلة سابقة

a. Elevated C-reactive protein
b. Decreased erythrocyte sedimentation rate
c. Elevated WBC count
d. Decreased hemoglobin and hematocrit

48 According to the Jones criteria, which sets of findings would indicate a positive diagnosis of rheumatic fever? (Select all that apply) أسئلة سابقة

a. Carditis, fever, and an elevated WBC count
b. Positive strep throat culture, arthritis, chorea
c. Recent scarlet fever, carditis, fever, arthralgia
d. Elevated C-reactive protein, carditis, fever

49 Acute Rheumatic Fever (ARF) is primarily caused by which bacterial species? أسئلة سابقة

a. Staphylococcus aureus
b. Streptococcus pneumoniae
c. Streptococcus agalactiae
d. Streptococcus pyogenes (Group A Streptococcus)

50 Molecular mimicry in Acute Rheumatic Fever refers to: أسئلة سابقة

a. The ability of bacteria to mimic host cells.
b. The interaction between antibodies and antigens.
c. The cross-reactivity between bacterial antigens and host tissues.
d. The production of toxins by Streptococcus pyogenes.

51 Which of the following clinical manifestations is NOT a major criterion for diagnosing Acute Rheumatic Fever? أسئلة سابقة

a. Carditis
b. Chorea
c. Elevated C-reactive protein
d. Subcutaneous nodules

52 Patient with polyarthritis and carditis with evident cardiomegaly is treated with أسئلة سابقة

a. IV salicylates
b. Oral salicylates
c. Corticosteroid
d. Adrenaline

53 Which of the following in not considered among supportive therapies for patient with carditis

أسئلة سابقة

a. Digoxin
b. Diuretics
c. Oxygen
d. Increasing fluid and salt intake

54 Which clinical manifestation of Acute Rheumatic Fever involves rapid, jerky movements and behavioral changes?

أسئلة سابقة

a. Carditis
b. Arthritis
c. Chorea
d. Subcutaneous nodules

55 Which genetic marker can influence an individual's susceptibility to developing Acute Rheumatic Fever after a streptococcal infection?

أسئلة سابقة

a. HLA-B27
b. Rh factor
c. ABO blood type
d. Genetic markers associated with the immune response

56 Which of the following is a major clinical manifestation of Acute Rheumatic Fever that involves inflammation of the heart?

أسئلة سابقة

a. Chorea
b. Arthritis
c. Carditis
d. Subcutaneous nodules

57 Which of the following is NOT a potential complication of Acute Rheumatic Fever?

أسئلة سابقة

a. Rheumatic heart disease
b. Glomerulonephritis
c. Meningitis
d. Chorea

58 Which of the following is a minor criterion for diagnosing Acute Rheumatic Fever?

أسئلة سابقة

a. Carditis
b. Arthritis
c. Elevated erythrocyte count
d. Subcutaneous nodules

59 What is the gold standard for diagnosing Group A Streptococcus infection in the laboratory?

أسئلة سابقة

a. Rapid strep test
b. Gram staining
c. Throat culture
d. Molecular PCR test

60 Which of the following medications is NOT commonly used in the treatment of Acute Rheumatic Fever?

أسئلة سابقة

a. Penicillin
b. Nonsteroidal anti-inflammatory drugs (NSAIDs)
c. Corticosteroids
d. Antibiotics targeting Gram-negative bacteria

61 Rheumatic fever primarily affects which of the following body systems?

أسئلة سابقة

a. Digestive system
b. Musculoskeletal system
c. Respiratory system
d. Nervous system

62 Which of the following is the characteristic feature of rheumatic fever? أسئلة سابقة

a. Joint pain and swelling
b. Skin rash
c. Heart murmurs
d. All of the above

63 Which of the following age groups are at higher risk for rheumatic fever? Select the correct answer أسئلة سابقة

a. Adults 50 and above
b. Children ages 5-15
c. Infants
d. Women ages 20-40

64 The best way to prevent rheumatic fever is: أسئلة سابقة

a. Avoiding contact with infected individuals
b. Getting vaccinated
c. Early diagnosis and treatment of streptococcal infections
d. Washing hands frequently

65 Rheumatic fever can lead to long-term complications, primarily affecting the..... أسئلة سابقة

a. Heart
b. Kidneys
c. Liver
d. Lungs

66 The condition characterized by scarring and damage to the heart valves as a consequence of rheumatic fever is called..... أسئلة سابقة

a. Coronary artery disease
b. Rheumatic heart disease
c. Rheumatoid arthritis
d. Septicemia

67 Which of the following nursing diagnoses is most appropriate for a child with rheumatic fever? ميدترم 2022 (أطفال نظري)

a. High risk for injury
b. Imbalanced nutrition: more than requirements
c. Delayed growth and development
d. Impaired gas exchange

68 A nurse is preparing a child for discharge after being diagnosed with rheumatic fever. Which is the correct instruction? ميدترم 2022 (أطفال نظري)

a. Only give penicillin for dental work
b. Give aspirin for management of chorea
c. Give penicillin until 21 years old
d. Give penicillin for one month total

69 Which of the following instructions would the nurse include in a teaching plan that focuses on prevention of rheumatic fever? ميدترم 2024 (ترم أول 2023-2024) (أطفال نظري)

a. Use corticosteroids to reduce inflammation
b. Provide antibiotic before dental work
c. Give penicillin to patients for treatment
d. Seek immediate treatment for throat infection

70 One minor manifestation of rheumatic fever is: ميدترم 2024 (ترم أول 2023-2024) (أطفال نظري)

a. Erythema marginatum
b. Polyarthritis and chorea
c. Increased sedimentation rate
d. Painless subcutaneous nodules

Part B · NCLEX-Style Practice

أسئلة على طريقة NCLEX (إضافة)

N1 Rheumatic fever is an autoimmune reaction that follows infection with: NCLEX

a. Staphylococcus aureus
b. Group A beta-hemolytic streptococcus
c. Pneumococcus
d. E. coli

N2 Select ALL MAJOR Jones criteria. SELECT ALL

a. Carditis
b. Polyarthritis
c. Chorea
d. Erythema marginatum
e. Subcutaneous nodules
f. Elevated ESR

N3 Secondary prophylaxis after rheumatic fever usually consists of: NCLEX

a. Oral aspirin for one month
b. Benzathine penicillin IM every 3–4 weeks for years
c. Corticosteroids for life
d. Antibiotics only during dental work

N4 A child with RF has chorea. The priority nursing diagnosis is: NCLEX

a. Impaired gas exchange
b. Risk for injury
c. Constipation
d. Excess fluid volume

N5 Which finding indicates carditis in a child with rheumatic fever? NCLEX

a. New heart murmur and tachycardia out of proportion to fever
b. Migratory joint pain
c. Ring-shaped rash
d. Painless nodules on extensor surfaces

N6 The joint involvement in rheumatic fever is typically: NCLEX

a. Small joints, symmetric and deforming
b. Migratory polyarthritis of large joints, non-deforming
c. Spinal only
d. Single joint with permanent damage

Part C · Quizlet Flashcards

كروت مراجعة سريعة

غطّي العمود الأبيض بورقة واختبر نفسك: شوف المصطلح وقول التعريف.

Cause	Autoimmune reaction 2–3 weeks after group A beta-hemolytic strep pharyngitis.
Molecular mimicry	Cross-reactivity between streptococcal antigens and host tissues.

Major Jones criteria	Carditis, polyarthritis, chorea, erythema marginatum, subcutaneous nodules.
Minor criteria	Fever, arthralgia, raised ESR/CRP, prolonged PR interval.
Evidence of strep	Positive throat culture (gold standard) or raised ASO titer.
Arthritis	Migratory, large joints, non-deforming.
Chorea (Sydenham)	Involuntary jerky movements, emotional lability; risk for injury.
Carditis	Only manifestation causing permanent damage → rheumatic heart disease (mitral).
Treatment	Penicillin, salicylates for arthritis, steroids for carditis with cardiomegaly/CHF, bed rest.
Secondary prophylaxis	Benzathine penicillin IM every 3–4 weeks (often until 21 years).
Primary prevention	Prompt treatment of strep throat.
Age group	Children 5–15 years.

Answer Key

الإجابات

1. b — An inflammatory disease following a bacterial infection

2. a — Streptococcus pyogenes

3. b — Fever, joint pain, and skin rash

4. a — Subcutaneous nodules

5. a — Throat swab culture

6. b — Preventing recurrence and complications

7. a — Valvular damage

8. b — Early treatment of streptococcal infections

9. a — Penicillin

10. b — Children aged 5-15 years

11. b — Cardiovascular system

12. c — Skin ulcers

13. a — Echocardiography

14. a — Fever

15. a — Uncontrollable jerky movements

16. a — Autoimmune cross-reactivity

17. b — Penicillin

18. a — Heart murmur

19. d — Until age 21
الملف معلّم بـ carditis: 5 سنين أو لحد سن 21 (الأطول منهم) until age 21.

20. a — True

21. b — False

22. b — False

23. b — False

24. a — True

25. c — 2–3 weeks
إجابة مقترحة (المصدر مفيهوش إجابة رسمية).

26. b — Knees, ankles, elbows and wrists
إجابة مقترحة (المصدر مفيهوش إجابة رسمية).

27. b — Pain that moves from joint to joint (migratory)
إجابة مقترحة (المصدر مفيهوش إجابة رسمية).

28. d — Headache
إجابة مقترحة (المصدر مفيهوش إجابة رسمية).

29. b — Hot or tender joints
إجابة مقترحة (المصدر مفيهوش إجابة رسمية).

30. b — Musculoskeletal, cardiovascular and nervous
إجابة مقترحة (المصدر مفيهوش إجابة رسمية).

31. b — Fever and joint pain
إجابة مقترحة (المصدر مفيهوش إجابة رسمية).

32. b — Erythema marginatum
إجابة مقترحة (المصدر مفيهوش إجابة رسمية).

33. c — Penicillin
إجابة مقترحة (المصدر مفيهوش إجابة رسمية).

34. d — Nausea
إجابة مقترحة (المصدر مفيهوش إجابة رسمية).

35. c — Increased sedimentation rate

36. c — Give penicillin until 21 years old

37. a — High risk for injury

38. b — False

39. c — Heart

40. c — Joint pain and fever

41. b — False

42. a — Acetaminophen

43. b — Oral salicylates

44. c — Changes in vision

45. c — Streptococcus

46. c — The mother's history of type-2 diabetes

47. a, c — Elevated C-reactive protein; Elevated WBC count

48. b, c — Positive strep throat culture, arthritis, chorea; Recent scarlet fever, carditis, fever, arthralgia

49. d — Streptococcus pyogenes (Group A Streptococcus)

50. c — The cross-reactivity between bacterial antigens and host tissues.

51. c — Elevated C-reactive protein

52. c — Corticosteroid

53. d — Increasing fluid and salt intake

54. c — Chorea

55. d — Genetic markers associated with the immune response

56. c — Carditis

57. c — Meningitis

58. b — Arthritis

59. c — Throat culture

60. d — Antibiotics targeting Gram-negative bacteria

61. b — Musculoskeletal system
إجابة البنك: musculoskeletal (المفاصل أشيع عرض). لكن أخطر وأدوم تأثير على القلب.

62. d — All of the above

63. b — Children ages 5-15

64. c — Early diagnosis and treatment of streptococcal infections

65. a — Heart

66. b — Rheumatic heart disease

67. a — High risk for injury

68. c — Give penicillin until 21 years old

69. d — Seek immediate treatment for throat infection

70. c — Increased sedimentation rate

N1. b — Group A beta-hemolytic streptococcus
Molecular mimicry between streptococcal antigens and host tissues (heart, joints, brain, skin) causes RF.

N2. a, b, c, d, e — Carditis; Polyarthritis; Chorea; Erythema marginatum; Subcutaneous nodules
The 5 major criteria are carditis, polyarthritis, chorea, erythema marginatum and subcutaneous nodules; elevated ESR/CRP is minor.

N3. b — Benzathine penicillin IM every 3–4 weeks for years
Long-acting penicillin prevents recurrence; duration depends on carditis (often until age 21 or longer).

N4. b — Risk for injury
Involuntary jerky movements put the child at risk of falls and injury — provide a safe, quiet environment.

N5. a — New heart murmur and tachycardia out of proportion to fever
Carditis presents with new murmurs, tachycardia, cardiomegaly or pericardial rub; it is the only manifestation that can cause permanent damage.

N6. b — Migratory polyarthritis of large joints, non-deforming
RF arthritis migrates between large joints (knees, ankles, elbows, wrists) and heals without deformity.